two-dimensional because you need two pieces of information—like latitude and longitude on the earth's surface—to specify a location). As with the circle, you should envision the sphere tacked on to every point of the usual dimensions, even though in Figure 12.8a, to keep the image clear, we draw only those that lie on the intersections of grid lines. In a universe of this sort, you would need a total of five pieces of information to locate a position in space: three pieces to locate your position in the big dimensions (street, cross street, floor number) and two pieces to locate your position on the sphere (latitude, longitude) tacked on at that point. Certainly, if the sphere's radius were tiny—billions of times smaller than an atom— the last two pieces of information wouldn't matter much for comparatively large beings like ourselves. Nevertheless, the extra dimension would be an integral part of the ultramicroscopic makeup of the spatial fabric. An ultramicroscopic worm would need all five pieces of information and, if we include time, it would need six pieces of information in order to show up at the right dinner party at the right time.

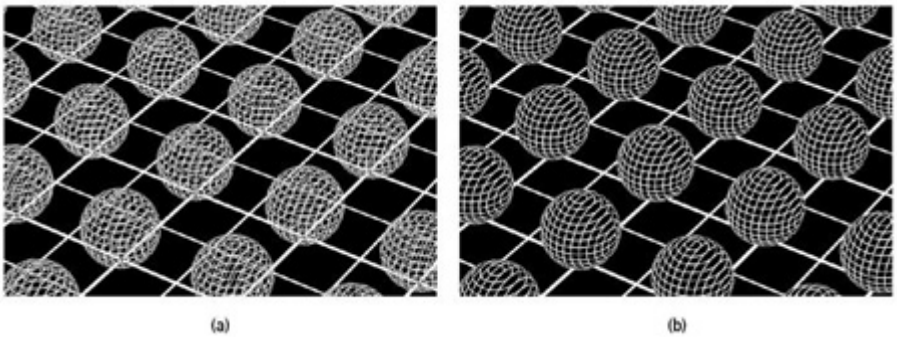


Figure 12.8 A close-up of a universe with the three usual dimensions, represented by the grid, and **(**a**)** two curled-up dimensions, in the form of hollow spheres, and **(**b**)** three curled-up dimensions in the form of solid balls.

Let's go one dimension further. In Figure 12.8a, we considered only the surface of the spheres. Imagine now that, as in Figure 12.8b, the fabric of space also includes the interior of the spheres—our little Planck-sized worm can burrow into the sphere, as ordinary worms do with apples, and freely move throughout its interior. To specify the worm's location would now require *six* pieces of information: three to locate its position in the usual extended spatial dimensions, and three more to locate its position in the ball tacked on to that point (latitude,